

The Effect of Digital Marketing on Customer Relationship Management: An Exploratory Study at Asiacell Telecommunication Company in Iraq

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Abstract—This paper aims to test the effect of digital marketing on customer relationship management in Asiacell Telecommunication Company. The Exploratory approach was used, and the questionnaire was adapted to collect data from the research sample that included (250) employees. The obtained data were analyzed by using (mean, standard deviation, relative importance, correlation coefficient, and simple linear regression) by relying on the statistical analysis program SPSS Ver.24. The mean result that the paper reached is that there is an effect from digital marketing on customer relationship management, and that means if there is any improvement in using digital marketing, that is enhance the level of customer relationship management in the researched company.

Keywords— Digital Marketing, Customer Relationship Management, Exploratory Study.

I. INTRODUCTION

From the mid-1990s until today, digital media has received significant attention. Since the creation of the Internet, the business world has changed significantly with the redefinition of many areas, such as marketing and customer relationships. This media creates the bath for conducting business through electronic tools; the growth of internet applications has stimulated many companies to create online stores and acquire many customers in most parts of the world [1]. Intense competition has forced organizations to rethink new marketing methods to reach customers faster and easier, so the need for modern and integrated marketing methods compatible with current modern technological changes has emerged [2]. Given the spread of commercial uses of the Internet and the opening of new horizons in the world of marketing, the importance of digital marketing has increased [3]. A development in marketing channels, such as distribution and communication, accompanied this development in marketing. This development contributed to finding new ways to communicate and interact with customers [4].

Therefore, organizations gradually began to realize the importance of digital marketing and customer relationship management in reaching new tools that contribute and adapt to develop and accelerate their response to the change in the business environment. Digital marketing is essential because it uses necessary capabilities and modern technological techniques to take appropriate measures. It has become an essential option or tool that helps companies reach the required goals, as digital

marketing methods allow companies to interact freely with customers and provide them with multiple services in a way that makes it easier for marketers to reach their customers and interact with them, which leads to improving the relationship with customers through participation and providing opportunities to increase the company's profits, and this requires building a long-term relationship with them.

II. RESEARCH METHODOLOGY

A. Research Problem

Given the growth of the communication sector in Iraq and the diversity of activities of various telecommunications companies in their quest to achieve excellence in digital marketing and how to manage customer relationships, the researchers noticed through their field visit to the researched company and their virtual visit to the company's website a notable interest from the company in adopting marketing activities. This motivated the researchers to test the impact of this trend towards digital marketing in Asia Cell Telecommunications Company in Iraq due to the existence of fertile ground and a suitable climate for applying the research and because it is one of the leading companies in Iraq in adopting digital marketing. Hence, the main research problem stated in this question is: "Does digital marketing affect customer relationship management in Asiacell Telecommunication Company?"

From this question, there are four sub-questions emerge, which can contribute to clarifying the research problem:

- Is Asiacell Telecommunication Company interested in digital marketing?
- Is Asiacell Telecommunication Company interested in customer relationship management?
- Is there a relationship between digital marketing and customer relationship management at Asiacell Telecommunication Company?
- Is digital marketing impacting customer relationship management at Asiacell Telecommunication Company?

B. Research Objectives

- Identifying the level of application of digital marketing in the researched company. Identifying the level of interest in

- b. customer relationship management in the researched company.
- c. Testing the relationship and effect between digital marketing and customer relationship management in Asiacell Telecommunication.

C. Data Collection Tools

The research adopted a descriptive-analytical approach in collecting the required data and information using a questionnaire form, which included the main variables: digital marketing, which includes seven dimensions, and customer relationship management, which also includes seven dimensions. The measures adopted in designing the questionnaire items are as follows: As for the independent variable (digital marketing), The study of [5] and [6] was adopted. As for the dependent variable (customer relationship management), the study of [7] and [8] were adopted.

D. Research model and hypotheses

The research model was designed to illustrate the form of relationship and influence between digital marketing and customer relationship management. Figure (1) illustrates the relationship and effects between the variables and their dimensions.

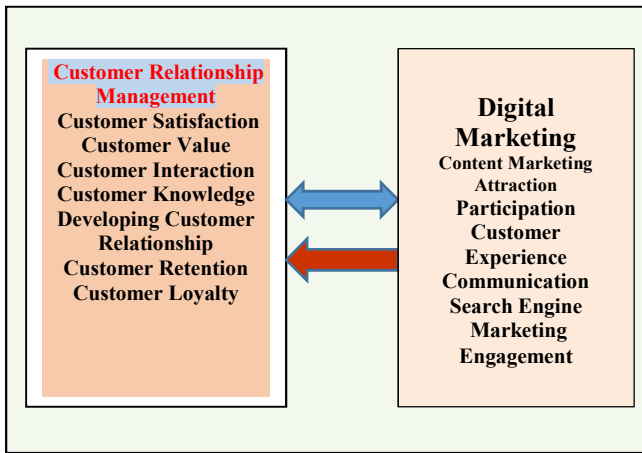


Fig 1. Research Diagram.

The first main hypothesis: There is a significant statistical correlation between digital marketing and customer relationship management.

The second main hypothesis: There is a significant statistical effect of digital marketing on customer relationship management.

E. Research's site and Population and Sample

The research site is "The AsiaCell Telecom Company/ Baghdad branch; the research population is the employees of the company, which are (590) employees, while the research sample is (250) employees. The questionnaire was sent electronically to the sample, and all the electronic questionnaires received were valid for statistical analysis. In order to test the hypotheses and show the causal relationships and effects between the variables. The obtained data were submitted to statistical analysis to accept

or reject the hypotheses, relying on (SPSS Ver.24). statistical program.

III. CONCEPT AND DEFINITION OF DIGITAL MARKETING

Digital marketing is considered a relatively new field, as there are many related things that companies need to continue learning and many modern concepts that need to be understood. Digital marketing is considered part of almost all main businesses, from product development and pricing to public relations and even recruitment [9]. Digital marketing is one of the modern concepts, and a new concept is added daily to keep pace with the tremendous technological developments. Organizations compete to provide more of it [10]. Digital marketing uses digital techniques that are measurable as they allow the organization to communicate and interact with customers by knowing complete details about them to facilitate the formation of a relationship between it and its customers.

In contrast, traditional marketing relies on broad communication directed to many customers, which results in difficulty in creating a relationship with customers quickly using fast means, as is the case using digital means [11]. Kotler defined digital marketing as "marketing products or services using digital techniques by Internet, including mobile phones, display advertising and any other digital Media" [12]. As for Kamal, he defined it as "a periodic practice carried out by commercial businesses, through analyzing customer needs, which is followed by manufacturing products and providing services, which also gives space for proper pricing and a study of the appropriate place where the sale process takes place, and the most important stage in this practice revolves around monitoring the satisfaction resulting from the customer after consuming the sold product or service" [13]. While Juma defined it as "marketing that is done by electronic technologies such as phones and computers so that stakeholders can be involved to be part of the organization's activities and operations, as digital marketing techniques enjoy their ability to replicate many aspects of marketing communications and old communication channels" [14].

IV. DIMENSIONS OF DIGITAL MARKETING

Table I shows the most important dimensions that researchers have adopted.

TABLE I. ADOPTED DIMENSIONS IN DIGITAL MARKETING

	Dimensions	Description
1	Content Marketing [15]	A marketing method that focuses on gaining and retaining customers by providing useful content to customers, which works to expand the scope of the purchasing experience and generate brand awareness.
2	Attraction [16]	The customer interaction requires them to visit the organization's interactive application, i.e., its website. It includes attracting customers to the site, building an audience, drawing attention to advertisements, and preserving the brand in memory. Marketers can attract customers through advertisements on websites or relevant links.
3	Participation [17]	Advanced practices carried out by the organization in customer service related to improving and developing customers' abilities to

		participate in developing activities, businesses, and services provided to its customers, such as participating in product design, marketing methods, and distribution. These practices can make customers active in production, design, marketing, and sales.
4	Customer Experience [18]	A mix of interactions between the customer and the organization, through which mutual value is acquired, and customers' response to any direct or indirect contact with it. It is the customer's cognitive and emotional evaluation of all direct and indirect encounters with the companies related to their purchasing behaviour.
5	Communication [19]	This means continuing to establish positive relationships with customers and knowing their needs and desires through communicating with them using traditional and digital marketing mix elements.
6	Search Engine Marketing [20]	All techniques used by search engines that develop a website's visibility by ranking it higher in search engine results are considered digital marketing. Digital marketing is one of the most potent forms of marketing, as it is the primary source of potential customers, bringing high website visit rates, reducing advertising costs, and building brand awareness and credibility.
7	Engagement [21]	Marketers make customers immersed and interactive by focusing on achieving interaction with the services provided. Digital marketing applications should be attractive and inspiring through their content to put customers in a state of interaction with these effects, creative programming techniques for advertising media, and providing valuable content to customers.

V. CUSTOMER RELATIONSHIP MANAGEMENT

A. Concept and Definition of Customer Relationship Management

The relationships between customers and organizations contribute to providing value to customers by offering information and receiving personalized service that meets their individual needs. This interaction forms the basis of a learning relationship, an ongoing and collaborative dialogue between the organization and the customer to grow more intelligently and interactively. Therefore, customer relationship management is a tool and strategy that helps the organization and its team stay on the right track to be able to anticipate the changing market landscape, whether customers are internal or external, communicating with the organization electronically or face-to-face, from different parts of the world or local. Customer relationship management is the path of organizations to success. [22] defined customer relationship management as "a strategy that includes human aspects and technological processes, which the organization follows to implement certain decisions and a set of targeted activities, including enhancing and developing relationships with customers and understanding their multiple needs and desires. Meanwhile, [23] defined it as "an ongoing process that involves developing and utilizing market information to build and maintain customer relationships."

B. Dimensions of Customer Relationship Management

Table II shows the most important dimensions adopted by the researchers.

TABLE II. ADOPTED DIMENSIONS IN CUSTOMER RELATIONSHIP MANAGEMENT

	Dimensions	Description
1	Customer Satisfaction [24]	It is the customers' reaction that appears through their use of a specific product or service based on their needs, after which it is determined whether the buyer will become a continuous customer or not
2	Customer Value [25]	Customer relationship management can influence the expected value for the customer through the organization's offers, and in short, customer relationship management that adopts a sound business strategy can improve customers' perception of the offer and meet the demand for customers' special needs
3	Customer Interaction [26]	It helps in improving and developing the relationship with the customer because it is one of the ways used by the organization to obtain information to meet customer needs, and also helps the organization to provide better products and services to the customer
4	Customer Knowledge [24]	It helps enable the organization to respond to their needs based on their purchasing behaviour and provide appropriate services to them, discover customer tastes and personalize products as a result of knowing customers and their preferences
5	Developing Customer Relationship [26]	The organization must know what customers like and dislike in order to strengthen and develop their relationship. The organization has a great opportunity to build long-term relationships with its customers because knowing their customers is considered a great competitive advantage for it
6	Customer Retention [27]	It means maintaining ongoing long-term business relationships with them, and customer retention can be viewed as representing the number of customers dealing with the organization at the end of the fiscal year expressed as a percentage
7	Customer Loyalty [28]	It is a strategy aimed at attracting customers and turning them into customers for the purchasing process due to their feeling of satisfaction and their affiliation to deal with, purchase, and practice the services of the same organization without other organizations. Loyalty is considered a firm commitment by the customer to repurchase their preferred product in the future

VI. THE PRACTICAL ASPECT: STATISTICAL DESCRIPTIVE ANALYSIS OF THE VARIABLES

A. The Independent Variable (Digital Marketing)

Table (4) illustrates the results of the descriptive statistical analysis of digital marketing and its dimensions, as well as the importance of each dimension to others.

TABLE III. MEANS, STANDARD DEVIATION, AND RELATIVE IMPORTANCE OF THE INDEPENDENT VARIABLE AND ITS DIMENSIONS

	The independent variable and its dimensions	Mean	Standard deviation	Relative importance %	Arrange variables according to relative importance
X1	Content marketing	3.95	0.736	79	4
X2	Attraction	4.02	0.709	80.4	2
X3	Engagement	3.91	0.719	78.2	7
X4	Customer experience	3.97	0.699	79.4	3
X5	Communication	3.93	0.717	78.6	6
X6	Immersion	3.94	0.703	78.8	5

X7	Search engine marketing	4.09	0.620	81.8	1
X	Digital Marketing	3.97	0.700	79.4	

Where the above table indicates the following:

1. A high mean for the(digital marketing)variable, in general, amounting to (3.97), as well as for its dimensions, which confirms the existence of high interest from the company in digital marketing in all its dimensions covered by the research. The dimension of (search engine optimization) achieved the highest mean, while the dimension of (engagement) achieved the lowest.
2. The standard deviation for the digital marketing variable achieved a value of (0.700), which means there is a relative agreement among the research sample about the importance of digital marketing.
3. The relative importance of digital marketing dimensions indicated the superiority of the (search engine marketing) dimension in comparison with other digital marketing dimensions, as it came in first place with relative importance value (81.8%), while (the engagement dimension) achieved the lowest level of importance in compared with other dimensions, with relative importance value (78.2%), which indicates less agreement with this dimension from the research sample.

B. The dependent Variable (Digital Marketing)

Table IV. illustrates the results of the descriptive statistical analysis of customer relationship management, its dimensions, and the importance of each dimension concerning others.

TABLE IV. MEANS, STANDARD DEVIATION, AND RELATIVE IMPORTANCE OF THE INDEPENDENT VARIABLE AND ITS DIMENSIONS

	The dependent variable and its dimensions	Mean	Standard deviation	Relative importance %	Arrange variables according to relative importance
Y1	Customer satisfaction	4.06	0.642	81.2	2
Y2	Customer value	4.00	0.666	80	5
Y3	Customer interaction	3.94	0.717	78.8	6
Y4	Customer knowledge	3.90	0.683	78	7

TABLE V. CORRELATION COEFFICIENTS BETWEEN RESEARCH VARIABLES

Independent Variable \ Dependent Variable	Digital Marketing	Content marketing	Attraction	Engagement	Customer experience	Communication	Immersion	Search engine marketing
Customer Relationship Management	**0.897	**0.827	**0.839	**0.827	**0.815	**0.883	**0.769	0.770**

**Represents a high significant value with a confidence level of (99%)

Table (6) indicates that the results of testing the first main hypothesis, which assumes (The relationship between digital marketing and customer relationship management) showed a highly significant correlation with a value of (0.897) with a confidence level of (99%). That indicates a strong positive

Y5	Customer relationship development	4.10	0.636	82	1
Y6	Customer retention	4.05	0.664	81	4
Y7	Customer loyalty	4.05	0.660	81	3
Y	Customer Relationship Management	4.01	0.667	80.2	

Table(6) indicates the following :

1. A high mean for (the customer relationship management) variable in general, with a value (of 4.01), as well as for its dimensions, confirms the high interest from the surveyed company in customer relationship management in all its dimensions addressed by the research. The dimension of (relationship development with the customer) achieved the highest mean, while the dimension of (customer knowledge) achieved the lowest mean,
2. The standard deviation of customer relationship management achieved a value of (0.667), which means that the research sample is relatively agreeable about its importance.
3. The relative importance indicated the superiority of the (developing the relationship with the customer) dimension compared to other customer relationship management dimensions, as it came in first place with a relative importance value (82%). In contrast, the (customer knowledge) dimension achieved the lowest level of importance compared to other dimensions, with a relative importance value (of 78%). This indicates less agreement with this dimension from the research sample than other dimensions of the customer relationship management variable.

VII. TESTING HYPOTHESES OF CORRELATION AND EFFECT RELATIONSHIPS BETWEEN RESEARCH VARIABLES

A. Testing hypotheses of correlation relationship between research variables

Table (6) shows the results of (the Spearman test) for the correlation relationship between digital marketing and its dimensions and customer relationship management.

correlation between the two variables. This means accepting the first central hypothesis.

B. Testing the hypotheses of the effect of digital marketing on customer relationship management

Table (7) shows the simple linear regression test results for the second central hypothesis.

TABLE VI. THE RESULTS OF THE SIMPLE LINEAR REGRESSION TEST

Independent Variable Dependent Variable	Coefficient of determination R ²	Regression coefficient (Effect) β	Constant α	Calculated t-value	Calculated F-value	Significance level P	Nature of the relationship
Content marketing	0.684	0.608	1.628	**10.22	**104.45	0.000	High significant
Attraction	0.704	0.620	1.527	**9.212	**84.872	0.000	High significant
Engagement	0.684	0.734	1.149	**11.49	**132.16	0.000	High significant
Customer experience	0.664	0.775	0.942	**17.29	**298.94	0.000	High significant
Communication	0.694	0.745	1.089	**13.13	**172.31	0.000	High significant
Immersion	0.591	0.713	1.211	**13.43	**180.36	0.000	High significant
Search engine marketing	0.593	0.844	0.565	**13.55	**183.75	0.000	High significant
Digital Marketing effect hypotheses.	0.805	1.002	0.039	**22.96	**527.21	0.000	High significant
The validity of the second main hypothesis and its sub-hypotheses at a 99% confidence level.							

**It represents a highly significant value with a confidence level of (99%)

Table (7) indicates that the results of testing the second central hypothesis, which Assume (there is an effect from digital marketing on customer relationship management) indicated that the contribution rate of digital marketing in supporting customer relationship management is (80.5%), and the (20%) is because of the effect of other factors not tested in the current study model.

The value of the regression coefficient β (indicates that if the interest in digital marketing technical increases in the company by one unit, that leads to an increase in the level of customer relationship management by (100%), while the calculated values of (t) and (F) were (22.96) and (527.21) respectively, which are more significant than their tabular values (2.326) and (6.85) respectively, confirming the existence of a significant effect from digital marketing on customer relationship management with a confidence level of (99%).

VIII. CONCLUSIONS AND RECOMMENDATIONS

A. Conclusions

- There is great interest from the company (research site) in digital marketing and its dimensions (Content, Marketing Attraction, Participation, Customer Experience, Communication, Search Engine, Marketing, and Engagement)
- There is a great interest from the company (research site) in digital marketing and its dimensions (Customer Satisfaction, Customer value, Customer Interaction, Customer Knowledge, Developing Customer Relationships, Customer Retention, and Customer Loyalty)
- Accepting the first and second hypotheses, which state that there is a relationship and effect between digital marketing and customer relationship management.

B. Recommendations

The following recommendations were set for the researched company:

- Enhance the company's interest in digital marketing dimensions (research site).

- Enhance the interest in customer relationship management dimensions from the company (research site).
- Enhance digital marketing interest because it improves the company's customer relationship management. This is done by understanding it well and clarifying its objectives to all managers and department heads so that they can apply its dimensions correctly.

REFERENCES

- V. Anandha Valli, "Consumer Attitude towards Digital Marketing," International Journal of Science and Research (IJSR), vol. 13, no. 4, pp. 894–897, Apr. 2024, doi: 10.21275/sr24409195321.
- B. Paul, R. Sara, and A. Paolo, "Marketing Principles and Practice," Marketing, Mar. 2022, doi: 10.1093/hebz/9780192893468.003.0002.
- S. Purwar, "Digital Marketing: An Effective Tool of Fashion Marketing," SSRN Electronic Journal, vol. 3, no. 4, 2019, doi: https://doi.org/10.2139/ssrn.3318992.
- K. H. Kim and H. Moon, "Innovative digital marketing management in B2B markets," Industrial Marketing Management, vol. 95, pp. 1–4, May 2021, doi: https://doi.org/10.1016/j.indmarman.2021.01.016.
- M. A. J. Al-Shammari, "Digital marketing and its role in enhancing the mental image of the organization: an exploratory study for the employees of Asiaccell for Iraqi Communications", the Holy Karbala branch, Master's thesis, College of Administration and Economics, University of Karbala, Karbala, Vol. 13, no. 54, (2017), pp. 1- 35.
- D. J.M. Al-Madhi, "Digital Marketing and its Impact on Service Quality", an exploratory study of the opinions of a sample of managers in Zain Iraq for mobile communications, Master's thesis, College of Administration and Economics, Iraqi University, Baghdad, (2019).
- M. R. Musheib, "The Role of Customer Relations Management in Achieving Competitive Advantage", an analytical and descriptive study of a sample of workers in public hospitals in Baghdad, Master's thesis, College of Administration and Economics, University of Baghdad, Baghdad, (2011).
- Th. Al-Bakri, & A. H. Talib, "Managing Customer Relations and Marketing Performance", Arabic Edition, Dar Amjad for Publishing and Distribution, Amman, (2015).
- S. Kingsnorth, "Digital Marketing Strategy: An integrated approach to online marketing," led, Kogan Page Limited, Britain and United States, (2016).
- S. Rob, "E- Marketing :The essential guide to marketing in a digital world", 5 th , Edition , Quirk,(2013).
- Shihab, M. A., Marhoon, H. A., Ahmed, S. R., Radhi, A. D., & Sekhar, R. (2024). Towards Resilient Machine Learning Models: Addressing

- Adversarial Attacks in Wireless Sensor Network. *Journal of Robotics and Control (JRC)*, 5(5), 1599-1617.
- [12] M. Abdmajeed Shihab, M. Khalaf Hussein, N. Saud Abd, and S. Rashid Ahmed, "Towards Safer Roads: An Automated Traffic Control System with Enhanced Vehicle Speed Detection and Measurement Algorithm," *International Journal of Computing and Digital Systems*, vol. 17, no. 1, pp. 1–17, Jan. 2025, doi: 10.12785/ijcds/1571039701.
- [13] Hussain, A. S. T., Hakim, B. A., Ahmed, S. R., Abed, T. H., Taha, T. A., Hasan, T. S., ... & Tawfeq, J. F. (2024). Unlocking solar potential: advancements in automated solar tracking systems for enhanced energy utilization. *Journal of Robotics and Control (JRC)*, 5(4), 1018-1027..
- [14] R.Th. Juma, "The Impact of Digital Marketing on the Intent to Buy from International Fashion Brands: an analytical and descriptive study on all consumers of international fashion brands in Amman", Master's thesis, College of Business, Middle East University, Amman, (2021).
- [15] A. Saha, D. Kar, "Digital marketing during a pandemic: A study on impact of digital marketing on consumer Perception during lockdown in Guwahati city," *International Journal of Commerce and Management Research*, Vol.(6), No.(4), (2020), pp.(18-22).
- [16] M. A. Muhammad, "The Impact of Digital Marketing on the Perceived Value of the Customer", An Applied Study on Egyptian Airlines, *Journal of Trade and Finance*, No. (1), Vol. (40), (2020), DOI:10.21608/CAF.2020.125688
- [17] P. Kotler, "Marketing Management", 21th edition, Prentice-Hall, U.S.A, (2018).
- [18] C.I. Mbama, C. Ikechukwu, P. Ezepue, L. Alboul, & M. Beer, "Digital banking, customer experience and financial performance UK bank managers, perceptions", *Journal of Research in Interactive Marketing*, Vol.(12), No. (4), 2018, pp.(432- 451), Doi: <https://doi.org/10.1108/ijbm-11-2016-0181>.
- [19] Th. Y. Al- Bakri, "Marketing Communications and Promotion", First Edition, Dar Al Hamed, Amman, Jordan , (2015).
- [20] F. Filip, "The impact of digital marketing on company's business development in Croatia", thesis master in Managerial Informatics, Faculty of Economics and Business, University of Zagreb, (2020), <https://urn.nsk.hr/urn:nbn:hr:148:934080>
- [21] Aljohani, A. (2023). Zero-Trust Architecture: Implementing and Evaluating Security Measures in Modern Enterprise Networks. SHIFRA, 2023, 60-72.
- [22] L.E Mendoza, A. Marius, M. Perez, and A. C. Griman, "Critical Success Factors For A Customer Relationship Management Strategy" *Information and Software Technology*, Vol.49, No.8, 2017, PP.913-945, doi: <https://doi.org/10.1016/j.infsof.2006.10.003>.
- [23] M.K. Shulka, & P.N. Pattnaik, "Managing Customer Relations in a Modern Business Environment: Towards an Ecosystem-Based Sustainable CRM Model," *Journal of Relationship Marketing*, Vol.(18), No.(1), 2018, pp.(17- 33) , doi: <https://doi.org/10.1080/15332667.2018.1534057>.
- [25] V. Kumar, & W. Reinarts, "Customer Relationship Management: Concept, Strategy, and Tools", 3ed, Springer-Verlag GmbH, Germany, (2018), DOI: <https://doi.org/10.1007/978-3-662-55381-7>.
- [26] H. Al-Taie, & N.I. Abu Al-Khair, "E-marketing in the service sector, a conceptual and applied approach", first edition, knowledge efficiency for publishing and distribution, Amman, (2022).
- [27] F. Buttle, & S. Maklan, "Customer Relationship Management: Concepts And Technologies", 3ed, Routledge, New York, (2015), DOI: <https://doi.org/10.4324/9781351016551>.
- [28] R.Y Ibrahim, & M. Alazzawi, "The effect of service quality on managing customer relations, an applied study in a sample of Iraqi banks" *Journal of Administration and Economics*, V.(45), N. (124), 2020, p.p(62-81), DOI: <http://doi.org/10.31272/JAE.43.2020.124.5>